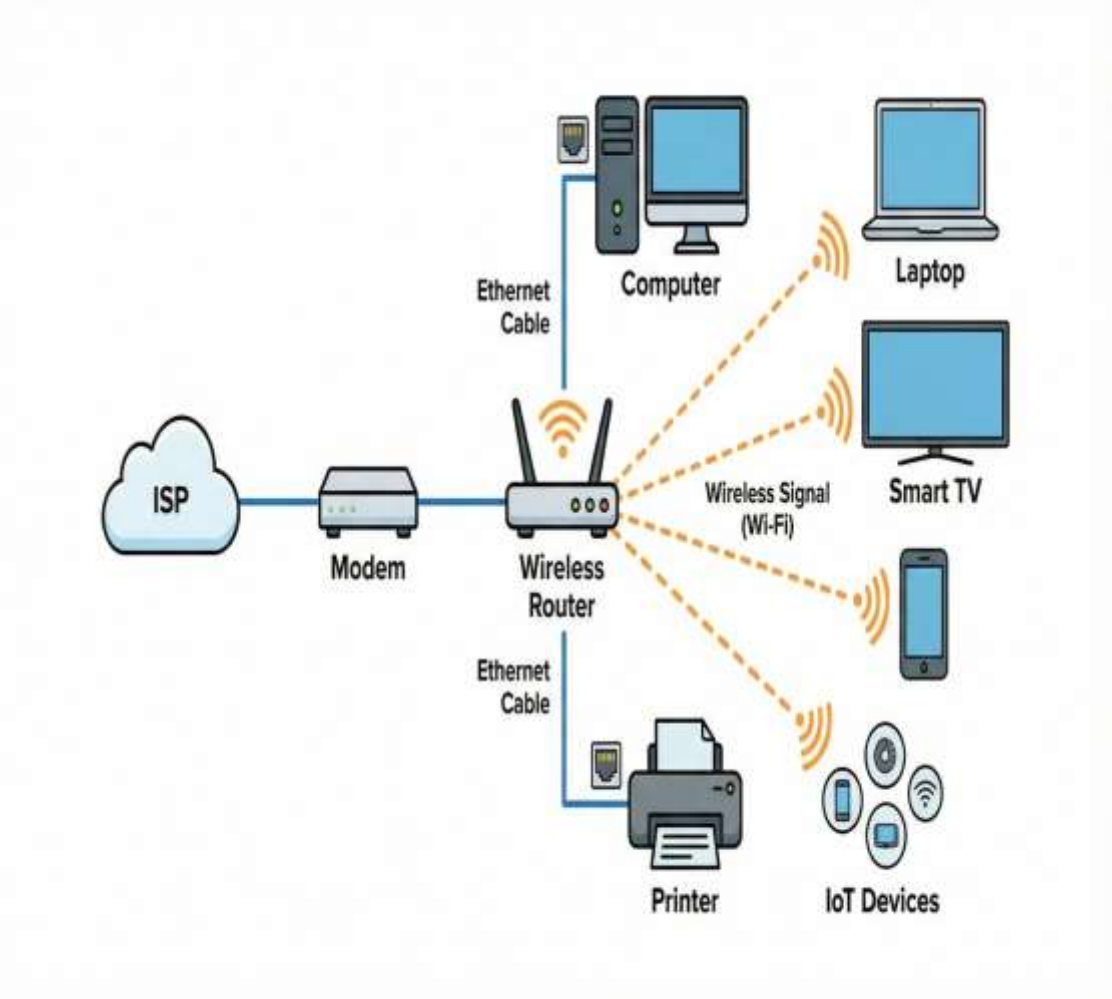
Day-2 routers and switches theory

Task-1 Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



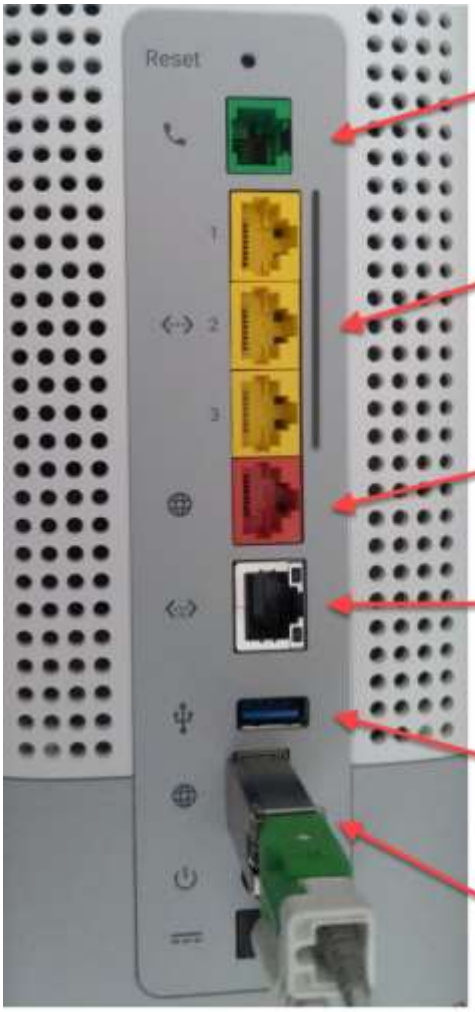
Day-2 routers and switches theory

Task-2 List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Feature	Router	Switch
Main Function	Connects different networks, such as a home LAN to the Internet.	Connects multiple devices within the same LAN.
Address Used	Mainly uses **IP addresses** to determine where packets should be routed.	Mainly uses **MAC addresses** to forward Ethernet frames.
Traffic / Networks	Routes traffic between different networks and can provide services such as NAT and DHCP.	Forwards traffic between devices on the same local network.

Day-2 routers and switches theory

Task-3 Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Green RJ-11 port with phone handset icon: Line for VOIP phone

3 yellow RJ-45 ports with arrow icons: 1 gbps LAN ports. (Ports intended for devices connected to LAN, i.e. PCs, etc)

Red RJ-45 port with globe icon: Port intended where direct ONT fiber not available or setup for Bridge mode. (Not intended for LAN connections)

Silver/copper RJ-45 port with the text "10" and arrow icons: 10gbps LAN port. (Port intended for a multi-gig switch or Wifi Mesh Tri-Band Extender)

Blue USB port: Intended for tech-support/troubleshooting service. Google Support stated there is no network connectivity supported over USB.

Silver SFP+ port with globe icon: 10gbps SFP port for WAN uplink. Google Support could not provide if this was an SFP or SFP+ port, but it's probably SFP+. (Port intended as uplink using fiber SFP module for connection to ONT.)

Day-2 routers and switches theory

Task-4 Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

First, I would connect the **ISP line to the router's WAN/Internet port**. The router would provide Internet access and manage communication between the café's local network and the ISP.

Next, I would connect one of the router's **LAN ports to the network switch** using an Ethernet cable.

Why this setup?

The **router** should be connected to the ISP because its job is to connect the café's local network to the Internet and route IP traffic.

The **switch** is then used to connect the 8 gaming PCs because a switch is designed to connect many devices efficiently within the same LAN. Using Ethernet connections also gives gaming PCs a more stable and generally lower-latency connection than relying on Wi-Fi.

Final setup:
ISP → Router → Switch → 8 Gaming PCs